

Tyrell Freeman

Cybersecurity Graduate Student | Governance, Risk & Compliance
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PROFESSIONAL SUMMARY

Cybersecurity graduate student at St. John's University with hands-on academic experience in risk analysis, security control mapping, threat intelligence, and cloud-based incident remediation. Applied NIST Cybersecurity Framework, NIST SP 800-53, OWASP, and CIS Controls across coursework in secure scripting, cryptography, and AWS security operations. Background in vendor management and operational logistics brings a practical perspective to governance and third-party risk. Seeking a GRC internship to apply framework knowledge and analytical skills in a professional environment.

CORE SKILLS & COMPETENCIES

Frameworks:	NIST CSF, NIST SP 800-53 (SC-12, SC-8), ISO 27001, OWASP Top 10, CIS Controls, MITRE ATT&CK, CISA Secure by Design
GRC Competencies:	Security control mapping, risk assessment, threat analysis, incident remediation, IOC documentation, audit trail review
Cloud & Tools:	AWS CloudTrail, CloudWatch, EventBridge, Config, Lambda, IAM, VPC, SNS; Wireshark; Python (cryptography library); PowerShell
Documentation:	Lab reporting, technical writing, policy-aligned procedure documentation, executive summaries, APA/IEEE citation
Regulatory Awareness:	HIPAA, CISA advisories, FBI/HHS joint cybersecurity guidance, PKI certificate standards (X.509, DV/EV)

ACADEMIC PROJECTS

AWS Cloud Incident Detection & Automated Remediation

Module 13 Learning Activity | St. John's University | Spring 2026

Configured and analyzed two AWS security labs covering automated incident detection and cloud-based remediation workflows. Lab 6.1 built a detection pipeline using CloudTrail, CloudWatch, and EventBridge to identify failed login attempts and unauthorized security group changes, with SNS email alerts for analyst notification. Lab 7.1 extended this to full automated remediation, where an AWS Config rule triggered a Lambda function to immediately revoke unauthorized inbound rules from EC2 security groups without human intervention. Applied the NIST Cybersecurity Framework across all five functions to analyze the security architecture of both labs.

- Mapped both lab architectures to NIST CSF: CloudTrail and AWS Config fulfilled Identify; IAM least-privilege roles fulfilled Protect; CloudWatch alarms and EventBridge rules fulfilled Detect; SNS notifications and Lambda remediation fulfilled Respond; Config timeline and CloudWatch logs fulfilled Recover.
- Simulated a misconfiguration incident by adding overly permissive inbound rules (HTTP, SMTPS, HTTPS, IMAPS open to 0.0.0.0/0) and verified that Lambda automatically revoked all unauthorized rules within minutes of detection.
- Reviewed IAM role scoping for AwsConfigRole and AwsConfigLambdaSGRole, confirming least-privilege design — each role was limited to only the permissions required for its specific function.
- Documented full audit trails including Config timelines, CloudWatch Logs Insights queries, and EventBridge event patterns used for detection and verification.

Tools / Frameworks: AWS CloudTrail, CloudWatch, EventBridge, Config, Lambda, IAM, VPC, SNS; NIST CSF

Cryptographic Key Distribution & Management — PKI Security Analysis

Module 12 Assessment | CYB 625: Principles in Secure Scripting & Cryptography | St. John's University

Group research report analyzing cryptographic key distribution and Public Key Infrastructure (PKI) in enterprise environments. Examined how CA trust hierarchies enable authentication at scale, mapped controls to NIST SP 800-53 standards, and demonstrated scripted approaches to automating certificate validation and CSR generation.

- Mapped PKI implementation practices to NIST SP 800-53 controls SC-12 (Cryptographic Key Establishment and Management) and SC-8 (Transmission Confidentiality and Integrity), documenting how scripted automation supports compliance with each control's requirements.
- Analyzed certificate validation tiers (DV, OV, EV) and their appropriate use cases, demonstrating how organizations match validation level to application risk — a direct application of risk-proportionate control selection.
- Scripted X.509 certificate parsing in Python to automate detection of fraudulent certificates and Unicode phishing indicators, reducing reliance on manual certificate review.
- Identified the Unicode homograph attack vector as a gap in CA validation processes and recommended enhanced monitoring controls to address the human-error component of PKI trust.

Tools / Frameworks: Python (cryptography library), PowerShell, NIST SP 800-53 (SC-12, SC-8), X.509 PKI

RansomHub Ransomware Threat Analysis — Risk & IOC Assessment

Module 12 Learning Activity: Malware in the News | St. John's University | April 2026

Threat analysis of RansomHub, a Ransomware-as-a-Service operation that compromised over 210 organizations across healthcare, government, water utilities, and financial services within months of its February 2024 emergence. Analysis was grounded in the joint advisory issued by the FBI, CISA, and HHS (AA24-242A) and examined the group's attack lifecycle, indicators of compromise, and organizational impact from the perspective of a threat detection team member.

- Documented network-based IOCs including C2 traffic over port 443 (HTTPS), lateral movement via RDP (3389) and SMB (445), Tor-based .onion communications, and data exfiltration using MEGAsync and WinSCP — providing a detection checklist aligned with SIEM alert configuration.
- Analyzed three actively exploited CVEs used for initial access: CVE-2023-3519 (Citrix NetScaler RCE), CVE-2023-22515 (Atlassian Confluence privilege escalation), and CVE-2020-1472 ZeroLogon — highlighting the risk of unpatched vulnerabilities in enterprise environments.
- Identified persistence mechanisms including scheduled task masquerading, HKCU registry run key modifications, and abuse of legitimate RMM tools (AnyDesk, Atera) — demonstrating how living-off-the-land techniques evade signature-based detection.
- Developed detection recommendations for SIEM and EDR platforms, including alerts for VSS deletion commands, abnormal PowerShell/WMI usage, and mass file encryption events, tied directly to the double-extortion kill chain.

Tools / Frameworks: CISA Advisory AA24-242A, SIEM/EDR detection concepts, CVE database, Tor/RDP/SMB analysis

Secure Scripting Techniques — Framework-Aligned Presentation

Intrusion Detection & Analysis | St. John's University

Developed and delivered a 12-slide technical presentation on mitigating security risks through secure scripting practices. The presentation applied multiple authoritative security frameworks to scripting vulnerabilities and defensive countermeasures, targeting an audience of cybersecurity practitioners and students.

- Referenced and applied six authoritative frameworks: NIST SP 800-53, NIST SP 800-218 (Secure Software Development Framework), OWASP Top 10, CIS Benchmarks, MITRE ATT&CK T1059 (Command and Scripting Interpreter), and CISA Secure by Design — demonstrating ability to navigate and synthesize GRC-relevant standards.
- Quantified the business risk of insecure scripting using the IBM Cost of a Data Breach 2024 report (\$4.88M average cost) and Verizon DBIR, establishing risk context before presenting technical controls.
- Covered six vulnerability categories (SQL/command injection, XSS, insecure deserialization, hardcoded credentials, path traversal, TOCTOU race conditions) with corresponding Python, PowerShell, and shell hardening countermeasures.
- Validated all 14 references against public sources, correcting three outdated NIST CSRC URLs and verifying open accessibility of OWASP, MITRE, and CISA resources — reflecting the diligence expected in compliance documentation.

Tools / Frameworks: NIST SP 800-53 & 800-218, OWASP, CIS Benchmarks, MITRE ATT&CK, CISA Secure by Design

EDUCATION

St. John's University — Queens, NY

Master of Science in Cybersecurity

Expected December

2027

Relevant Coursework: Principles in Secure Scripting & Cryptography (CYB 625), Intrusion Detection & Analysis, Cloud Security, Network Security, Information Security Management

All projects completed as part of the M.S. Cybersecurity program at St. John's University. Full reports and presentation materials available upon request.